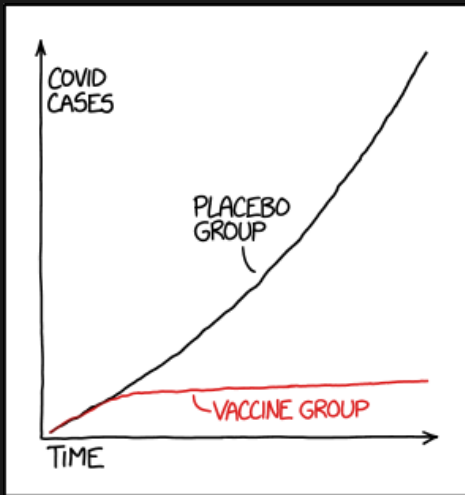


Towards Modern Statistical Testing

Daniel Lawson University of Bristol

Lecture 05.2 (v4.0.0)

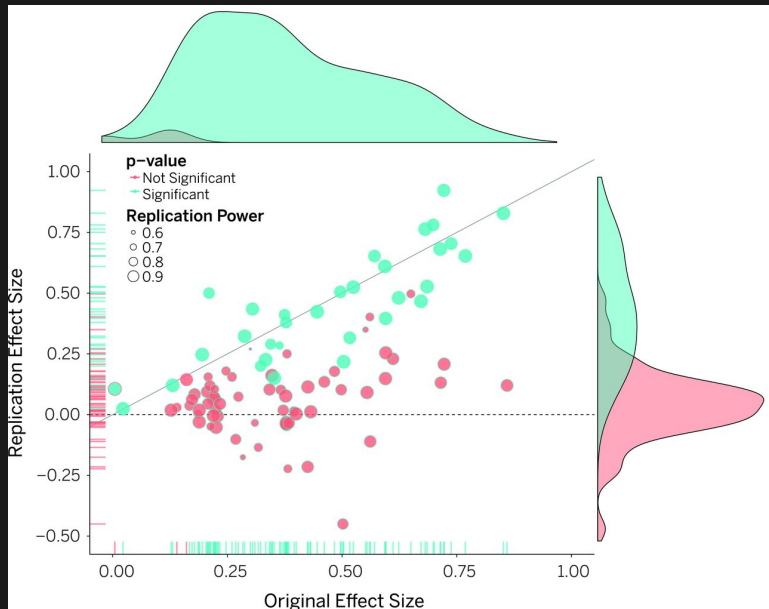


STATISTICS TIP: ALWAYS TRY TO GET DATA THAT'S GOOD ENOUGH THAT YOU DON'T NEED TO DO STATISTICS ON IT

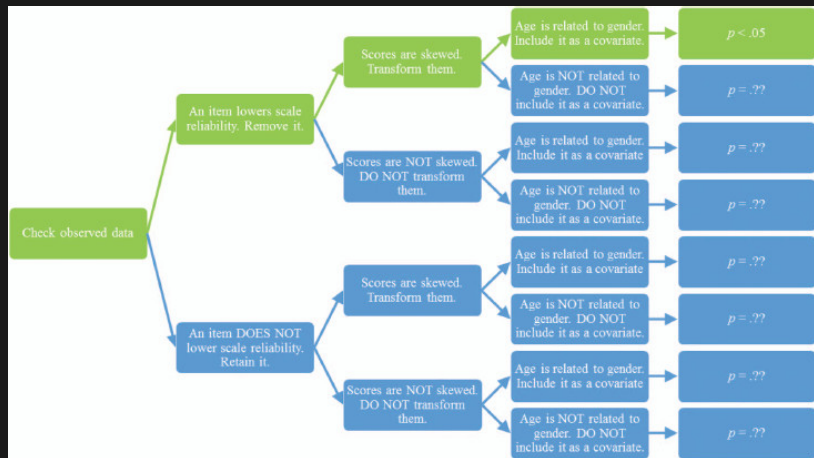
Signposting

- ▶ Classical statistics was all about p-values
- ▶ They recently started to become a lost art
- ▶ But modern machine learning needs rigorous uncertainty quantification more than ever

The replication crisis



The garden of forking paths



Rubin: Four Solutions to the Forking Paths Problem, Rev. Gen. Psych. 2017

Questions

- ▶ Why might statistical testing have a bad reputation?
- ▶ What is the point of testing in large-scale data science?
- ▶ What does(n't) Cross Validation save you from?

Null hypothesis test

- ▶ Given some data $\{y\}$:
 - ▶ Null Hypothesis **H0**: A statement is true about $\{y\}$.
 - ▶ Alternative Hypothesis **H1**: The statement is not true.
- ▶ We then compute a **test statistic** $T(\{y\})$ whose distribution is **computable under H0**.
 - ▶ By convention, large T is evidence against the null.
- ▶ Then compute p-value $p(T \geq T(\{y\}))$, the probability of observing a test statistic at least as large as that observed given H_0 is true.
 - ▶ Example: $H_0: \mathbb{E}(y) = \mu$ with $\mu = 0$. $H_1: \mu \neq 0$.
 - ▶ This is **not model selection**. We favour H_0 and must find evidence against it to accept H_1 .

Null hypothesis significance testing

- ▶ Hypothesis testing is asking: are my data consistent **with this hypothesis** when **using this measure**?
 - ▶ If you choose a silly hypothesis, testing will dutifully say “no”
 - ▶ If you use a weak measure, testing will dutifully say “yes”
 - ▶ Nothing is learned by this!
- ▶ The correct use of statistical testing is where:
 1. the **null hypothesis might plausibly be true**, or
 2. it might not be true, but you care how much **power the data has to reject the null**

When to use hypothesis testing

- ▶ Some valid use cases include:
 - ▶ To **rank hypotheses** by how much evidence there is against them
 - ▶ To obtain a **standardised scale** (0-1) for combining evidence
 - ▶ When **data are scarce**
- ▶ Also when testing plausible nulls, such as:
 - ▶ **validating simulations** with a known simulator;
 - ▶ **independence** or other non-parametric tests.
 - ▶ **broad null hypotheses**, such as testing a range of parameters.

Types of error

- ▶ The **p-value** defines *the probability that H_0 is true, but is rejected*.
- ▶ The **power of the test** is *the probability that H_0 is false but is accepted anyway*.
 - ▶ Low power situations are to be avoided: see e.g. Andrew Gelman's blog¹.
- ▶ Power is a surprisingly important problem because there are many *researcher degrees of freedom*.
 - ▶ so if power is low, we tend to find significant results anyway, through the (often unintentional) data and testing choice.

¹<https://andrewgelman.com/2018/02/18/low-power-replication-crisis-learned-since-2004-1984-1964/>

Types of error

Error notation

	H_0 true	H_0 false
H_0 accepted	Correct	Type II error
H_0 rejected	Type I error	Correct

Types of error

Error notation

	H_0 true	H_0 false
H_0 accepted	Correct	Type II error
H_0 rejected	Type I error	Correct

- Under the convention that $H_0 = 0$ = “negative” case and $H_1 = 1$ = “positive case”:

Alternative notation

	H_0 holds	H_1 holds
H_0 accepted	True Negative	False Negative
H_0 rejected	False Positive	True Positive

Classical tests

- ▶ t-tests, chi-squared test, etc make **parametric assumptions**
 - ▶ the data are normal, identical and independently distributed, etc
 - ▶ its sometimes approximately true, so useful for small data

(Re)sampling

- ▶ The main types of resampling tests include:
 - ▶ **jackknifing**, which is analysing subsets of data “leave-k-out” style, to estimate (variance of) parameter estimates
 - ▶ **bootstrapping**, which is resampling with replacement, to estimate (variance of) parameter estimates
 - ▶ **permutation**, which is resampling without replacement, to test a null hypothesis
 - ▶ **cross-validation**, which is analysing subsets of data to estimate out-of-sample prediction, for model performance
- ▶ Each of these methods can be applied to a wide variety of problems, and often requires thought to use appropriately.

Why sampling

- ▶ Asymptotically correct and distribution free. We only (!) have to assume **exchangeability**.
- ▶ Exchangeability of what?
 - ▶ what would be **equal if the null hypothesis is true**, and
 - ▶ would be **different if the alternative hypothesis is true**?
- ▶ The test is correct if we **maintain any true correlation structure**.
- ▶ For example, if the indices were originally correlated, permutation testing will fail.
 - ▶ as from e.g. a time-series.

Modern Machine learning testing

Monte-Carlo test

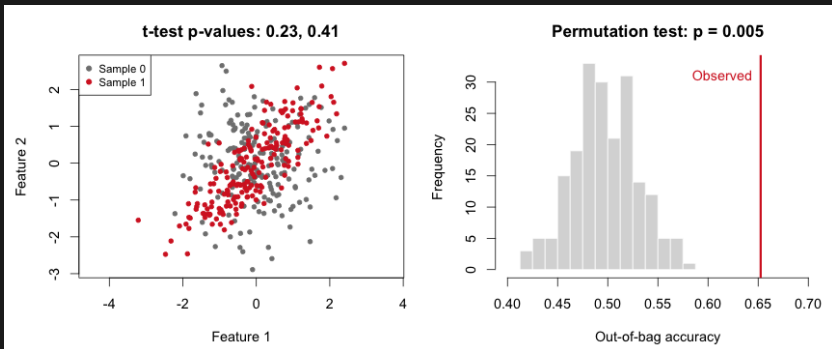
- ▶ To conduct a Monte-Carlo test, we construct N random datasets $\{x_i\}$ **under H_0** and add our real dataset X .
- ▶ We then ask, is the **real dataset an outlier** with respect to the random datasets?
- ▶ Specifically, the p-value for a test T applied to X (where large values are considered strange) is:

$$p = \frac{1 + \#\{i : T(x_i) \geq T(X)\}}{N + 1}$$

- ▶ This is **exact** for any N , and **any** statistic T , if X and the x_i are exchangeable under H_0 .
- ▶ So T can be anything we can compute, **including the output of a machine learning model**.

Example: a classifier as a test statistic

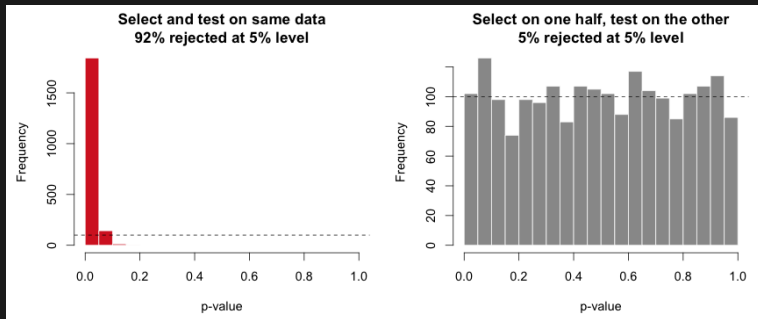
- ▶ **H0**: two samples come from the **same distribution**. If so, no classifier can tell them apart.
- ▶ T = out-of-sample accuracy of a RF classifier trained to separate them; permuting the labels imposes H0.



- ▶ Every feature has the same $N(0, 1)$ distribution in both samples; only the **correlation** differs.
- ▶ t-tests on each feature see nothing, but the forest does.

Testing after model selection

- ▶ ML pipelines **choose** features, models and hyperparameters using the data.
- ▶ Testing the chosen model **on the same data** gives invalid p-values (“double dipping”).
- ▶ Fixes: **split the data** (select on one part, test on the other), or put the **whole pipeline** inside the permutation loop.



- ▶ Null simulation: y is unrelated to all 50 features; we test the one most correlated with y .

Conformal prediction

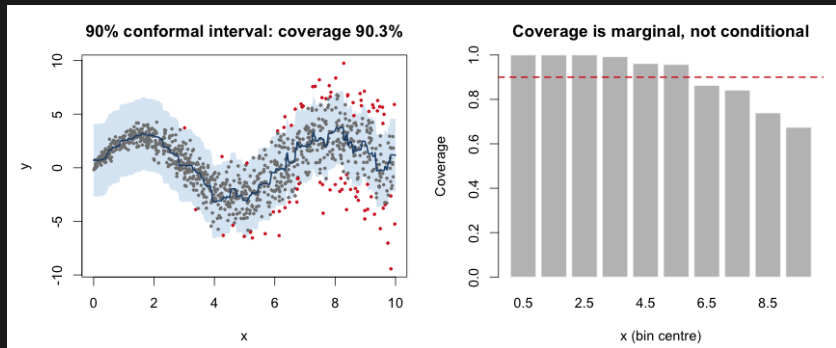
- ▶ **The idea**: wrap **any** model to get prediction intervals that contain the truth, say, 90% of the time.
- ▶ Hold out some **calibration data**, and look at how big the model's errors are there.
- ▶ Make the interval just wide enough to contain 90% of those errors.
- ▶ A new point's error is **equally likely to fall anywhere** in the ranking of calibration errors, so the interval covers it 90% of the time.
- ▶ This is the Monte-Carlo argument again. It assumes nothing about the model, only that new data are **exchangeable** with the calibration data.
- ▶ The details, and what “90% of the time” really means: Portfolio 5.2.P.

Example: conformal prediction intervals

```
alpha=0.1
## Split: fit on one half, calibrate on the other
idx=sample(nrow(dat))
tr=idx[1:500]; cal=idx[501:1000]
fit=randomForest(y~x,data=dat[tr,],nodesize=40)
## Conformity scores and the conformal quantile
s=abs(dat$y[cal]-predict(fit,dat[cal,]))
q=sort(s)[ceiling((length(cal)+1)*(1-alpha))]
## Prediction intervals for new data
pred=predict(fit,test)
interval=cbind(pred-q,pred+q)
```

- ▶ Any model can replace randomForest: only the scores s are needed.

Example: conformal prediction intervals



- ▶ The guarantee is **marginal** (on average over x), not **conditional** on x . Adaptive scores (e.g. conformalised quantile regression) do better.

Reflection

- ▶ You should understand how to:
 - ▶ Define and use a null hypothesis significance test,
 - ▶ Contrast classical and resampling tests, and judge appropriate uses,
 - ▶ Use a machine learning model as a Monte-Carlo test statistic, without double dipping,
 - ▶ Wrap any predictive model in conformal prediction intervals, and state what they guarantee,
 - ▶ Use statistical testing appropriately in projects.

Further reading

- ▶ Classical Testing
 - ▶ Chapter 4 of [Statistical Data Analysis by Glen Cowan](#)
 - ▶ [Statistical tests, P values, confidence intervals, and power: a guide to misinterpretations](#) by Greenland et al
 - ▶ [Andrew Gelman's blog](#) has many examples of statistical testing failures in social science and medicine

Further reading

▶ Modern Testing

- ▶ [Cosma Shalizi's Modern Regression Lectures](#) (Lectures 26,28)
- ▶ [Cross Validation and Bootstrap Aggregating](#) on Wikipedia
- ▶ Chapters 18.7 of [The Elements of Statistical Learning: Data Mining, Inference, and Prediction](#) (Friedman, Hastie and Tibshirani).
- ▶ [Revisiting Classifier Two-Sample Tests](#) by Lopez-Paz & Oquab (ICLR 2017), and [Permutation Tests for Studying Classifier Performance](#) by Ojala & Garriga (JMLR 2010)
- ▶ [A Gentle Introduction to Conformal Prediction](#) by Angelopoulos & Bates: the best starting point, with code

▶ Cross Validation

- ▶ Chapters 2.3 and 7.10 of [The Elements of Statistical Learning: Data Mining, Inference, and Prediction](#) (Friedman, Hastie and Tibshirani).
- ▶ [Cosma Shalizi's Modern Regression Lectures](#) (Lectures 20, 26)

Appendix: How many permutations?

- ▶ The **smallest possible p-value** with N permutations is $1/(N + 1)$. So 999 permutations gives a minimum of 0.001.
- ▶ The **variance** around a chosen threshold, say $p = 0.05$, is determined by the sampling distribution of the Binomial:

$$\text{sd}(p) = \text{sd}(\text{Bin}(N, p)) = \sqrt{\frac{p(1 - p)}{n}}$$

- ▶ p is of course the true unknown probability, not the observed one.
- ▶ But variance is an increasing function of p (for $p < 0.5$)
- ▶ A heuristic rule is: to be 95% confident that $p \leq t$ we need the empirical p-value to be less than $t - 1.96\text{sd}(p = t)$
- ▶ For $N = 999$ and $t = 0.05$, $\text{sd}(p = t) = 0.0135$ and therefore $p < 0.036$
- ▶ A similar calculation shows $N = 999$ wouldn't be enough to be sure we were less than 0.005.
- ▶ This is conservative... only if the distribution is Normal...(!)
Plot the distribution of T !

t-tests

- ▶ Can be one-tailed (**H0**: $\mu \leq \mu_0$) or two-tailed (**H0**: $\mu = \mu_0$)
- ▶ Assumes:
 - ▶ independence (note: paired tests are possible) and identically distributed
 - ▶ the **data are Normal**
 - ▶ the standard deviation is either known (t is then Normal) or estimated from the data (t is then t distributed).
- ▶ Used in regression, paired tests, etc.
- ▶ *NB Incomplete notes as this is a prerequisite!*

Chi squared test

- ▶ The χ^2 test is for categorical data comparing two variables.
- ▶ **H0**: No relationship between the variables; **H1** Some relationship between them.
- ▶ The **test statistic** for N datapoints from k classes, with x_i observations of type i , with expected value $m_i = Np_i$ where p_i is the expected probabilities, is (under the null):

$$X^2 = \sum_{i=1}^k \frac{(x_i - m_i)^2}{m_i} \sim \chi^2(k - 1)$$

- ▶ This is most often used for **contingency tables** though appears elsewhere.
- ▶ See also **Fishers exact test** for small samples.
- ▶ *NB Incomplete notes as this is a prerequisite!*

Other important tests

- ▶ Nonparametric tests:
 - ▶ **Mann-Whitney U or Wilcoxon rank sum** test: are two samples drawn from the same distribution? by comparing their ranks.
 - ▶ **Wilcoxon signed-rank** test - as rank sum test, for paired data.
 - ▶ **Kolmogorov-Smirnov** test - are two samples from the same distribution? by comparing the empirical cumulative distribution function.
- ▶ There are many online cookbooks which state exactly which circumstances each test should be used in. You should be able to use them.
- ▶ *NB Incomplete notes as this is a prerequisite!*